

Experiment 5: Complex Multi-Height + Multi-Wind + Obstacle | complex_mix | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1973	11.307	13.469	21.777	14.7%	27.9%	33.0%	7.947	17.644	5.305	10.483	0.408	0.953	1.319
PPO-GRU	1939	3.466	0.843	14.990	57.7%	76.0%	79.9%	9.445	18.255	6.578	14.767	0.381	0.888	1.189
PPO-RNN	1928	2.711	1.117	7.086	43.8%	71.6%	81.2%	14.188	26.817	7.015	15.844	0.445	1.211	1.472
MPC	959	4.765	2.618	13.709	3.0%	18.8%	69.7%	2.321	3.892	1.472	2.886	0.141	0.149	0.172

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1973	98.7%	27	1.4%	1.3%	26	96.3%	1	0
PPO-GRU	2000	1939	97.0%	61	3.0%	2.7%	53	86.9%	8	0
PPO-RNN	2000	1928	96.4%	72	3.6%	2.9%	58	80.6%	14	0
MPC	2000	959	48.0%	1041	52.0%	52.0%	1041	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	1041	1024	98.4%	715	69.8%	11.605
PPO-GRU	1041	988	94.9%	220	22.3%	3.771
PPO-RNN	1041	978	93.9%	215	22.0%	3.004